

Conclusions

*“Life can only be understood backwards
but it must be lived forwards”*

Soren Kierkegaard

In this chapter we discuss some final conclusions about the work presented in this thesis. In each of the previous chapters, we have already presented a *final considerations* section including a detailed discussion of the benefits of the approach proposed in that chapter, as well as its main advantages when compared to related work in the literature. For this reason, in the remaining of this chapter we focus mainly on two topics, namely, the most important research contributions of this thesis (sections 9.1 to 9.12), and plans for future work (section 9.13).

9.1 Language Evaluation and (Re)Design Framework

In chapter 2 of this thesis, we elaborate on the relation between a modeling language and a set of real-world phenomena that this language is supposed to represent. We focus on two aspects of this relation, namely, the *domain appropriateness*, i.e., the suitability of a language to model phenomena in a given domain, and its *comprehensibility appropriateness*, i.e., how easy is for a user of the language to recognize what that language’s constructs mean in terms of domain concepts and, how easy is to understand, communicate and reason with the specifications produced in that language. We defend that both these properties can be systematically evaluated for a modeling language w.r.t. a given domain in reality by comparing a concrete representation of the worldview underlying this language (captured in a *metamodel of the language*), with an explicit and

formal representation of a conceptualization of that domain, or a *reference ontology*.

We therefore propose a framework for language evaluation and (re)design which aims, in a methodological way, to approximate or to increase the level of homomorphism, between a metamodel of a language and a reference ontology. This framework comprises a number of properties (*lucidity, soundness, laconicity, completeness*) that must be reinforced for an isomorphism to take place between these two entities.

The framework proposed combines two existing proposals in the literature:

1. the one of Corin Gurr (Gurr, 1998, 1999), which focuses on the evaluation of individual representations;
2. the one of Yair Wand and Ron Weber (Wand & Weber, 1989, 1990; Weber, 1997), which aims at the evaluation of representation systems.

In addition, our framework extends these two proposals in several ways. Compared to them, our framework possesses the following advantages:

- Gurr uses regular algebraic structures to model a domain conceptualization. We strongly defend the idea that the more we know about a domain the better we can evaluate and (re)design a language for domain and comprehensibility appropriateness. As we show in chapter 2, there are important meta-properties of domain entities (e.g., *rigidity, relational dependency*) that are not captured by ontologically-neutral mathematical languages (such as algebras or standard set-theories), and that the failure to consider these meta-properties hinders the possibility of accounting for other important aspects in the design of visual modeling languages.
- In Wand & Weber's work, no attention at all is paid to pragmatic aspects of modeling languages. Here we show how the ontological meta-properties of the domain concepts captured in a reference ontology can be exploited in the design of efficient visual pragmatics for visual modeling languages.
- Wand & Weber's work focuses solely on the design of general conceptual modeling languages. The framework and the principles proposed here instead can be applied to the design of conceptual modeling languages irrespective to which generalization level they belong, i.e., it can be applied both at the level of material domains and corresponding domain-specific modeling languages, and at the (meta) level of a domain-independent (meta) conceptualization that underpins

a general conceptual (ontology) modeling language. In fact, this ontology-based framework amounts to an important contribution to the area of domain-specific languages design methodologies (as acknowledged, for instance, in Girardi & Serra, 2004).

- As in Wand & Weber’s proposal, the focus of our framework is on the level of systems of representations, i.e., on the evaluation of modeling languages, as opposed to a focus on individual diagrams produced using a language. Nevertheless, as it is shown in chapter 2, by considering desirable properties of the mapping of individual diagrams onto what they represent, we are able to account for desirable properties of the modeling languages used to produce these diagrams, extending in this aspect Wand & Weber’s work.

Wand & Weber’s framework addresses solely the relation between ontological categories and the modeling primitives of a language. It pays no attention to the possible constraints governing the relation between these categories. Moreover, it does not consider the necessary mapping from these constraints to equivalent ones, to be established between the language constructs representing these ontological categories. For example, consider two ontological categories A and B , which are represented in a language metamodel by two constructs C and D . If there are constraints on the admissible relations between A and B , these constraints must be also represented in the language metamodel in terms of the possible relations between C and D . In chapter 3, we refine our framework, by establishing a formal relation between the set of valid models that can be produced in a domain modeling language, and the set of state of affairs which are deemed admissible by a conceptualization of that domain. To put it simply: let O be an ontology of a domain conceptualization C such that $\mathcal{M}$ is a valid model of O iff it represents a state of affairs deemed admissible by C . Now, let S_L be the set of valid models of language $\mathcal{L}$ delimited by $\mathcal{L}$ ’s metamodel $\mathcal{MT}$; let S_O be the set of valid models of ontology O . We can formally define how truthful $\mathcal{L}$ is to the domain conceptualization C , by comparing how close the sets S_L and S_O are. In an ideal situation, these sets coincide and $\mathcal{MT}$ and O are isomorphic. In other words, in this case, $\mathcal{L}$ admits as valid models all and only those that represent admissible state of affairs according to C .

9.2 Ontological Foundations for Conceptual Modeling

The language evaluation and design framework proposed here can be applied both at the level of material domains (e.g., genomics, archeology,

multimedia, fishery, law, etc.) and corresponding domain-specific modeling languages, and at the (meta) level of a domain-independent (meta) conceptualization that underpins a general conceptual (ontology) modeling language. Nevertheless, due to the objectives of this thesis, we have focused on the level of meta-conceptualizations and of general conceptual modeling languages.

The domain appropriateness of a domain modeling language can be precisely defined in terms of the difference between the set of valid models of this language, and the set of *intended models* according to a given conceptualization of that domain. To put it baldly, an intended model according to a conceptualization C is a model that represents a state of affairs deemed admissible by C . Now, let C be a conceptualization of material domain in reality, such as genealogy. The possible state of affairs according to C would exclude, for instance, one in which a person is its own parent. As discussed in depth in chapter 3, the definition of which possible state of affairs comprises a material conceptualization C is far from arbitrary, but determined by the set of laws that constitute reality. So, we can define a meta-conceptualization C' , which contains all material conceptualizations C_i that are truthful to reality, i.e., domain conceptualizations that admit only state of affairs that can actually exist in reality. By applying the same line of reasoning to this meta-level of abstraction we have that a suitable general conceptual (ontology) modeling language that can be used to create domain ontologies (or conceptual models of material domains) is one which valid models are exactly the intended models of the meta-conceptualization C' .

Defining a repertoire of real-world categories (meta-conceptualization C') that can be used to articulate domain conceptualizations of reality is the very business of the discipline of *formal ontology* in philosophy. Thus, this task can greatly benefit from the conceptual tools and theories that have been developed along the years in this area. A meta-ontology representing a meta-conceptualization C' constructed using the theories developed by formal ontology in philosophy is named a *foundational ontology*.

From an ontological point of view, most conceptual modeling languages (e.g., UML, ER, LINGO, CCT, OWL) commit to a simplistic set-theoretical meta-conceptualization, and to an inadequate theory of universals (either *class* or *predicate nominalism*). These ontological choices can be warranted for the so-called *lightweight ontology representation languages* (e.g., the Semantic web languages), by arguing in favor of mathematical simplicity and convenience, or to justify practical performance trade-offs. However, as discussed in depth in chapter 6, there are many serious philosophical problems with these commitments, which make them unsuitable as a foundation for a general conceptual modeling language.

One of the main contributions of this thesis is to construct a meta-ontology which can be used as a foundation for conceptual modeling concepts and languages. As discussed in chapter 3, due to the objectives of conceptual modeling languages, this foundational ontology should be the result of a *descriptive metaphysics* effort. Thus, the conceptual modeling ontology proposed here considers not only results from metaphysics, but also from areas such as cognitive science and linguistics, with the main objective of capturing the ontological distinctions underlying natural language and human cognition.

The foundational ontology proposed here is one of the few and one of the most comprehensive approaches for developing *cognitive ontological foundations* for conceptual modeling languages found in the literature. The benefits of such a choice have been demonstrated in the related work sections of chapters 4 to 7 of this thesis.

The ontological foundations for conceptual modeling proposed here have been developed in four complementary parts, namely, classes and class hierarchies (chapter 4), part-whole relations (chapter 5), properties (chapter 6) and roles (chapter 7). In the following four subsections, we discuss the contributions of this thesis in each of these topics, respectively.

9.3 Theory of Universals and Universals Taxonomic Structures

In chapter 4, we present a well-founded theory of universals to address the topic of universals in conceptual (ontology) modeling. Universals are fundamental for conceptual modeling, being represented in all major conceptual modeling languages (e.g., OO classes, EER Entity types, OWL concepts, etc.), and in the practice of conceptual modeling, a set of concepts is often used to represent distinctions in different sorts of universals (Type, Role, State, Mixin, among others).

However, there is still a deficiency of methodological support for helping the user of the language to decide how to model elements that denote universal properties in a given domain and, hence, modeling choices are often made in ad hoc manner. Likewise it is the judgment of what are the admissible relations between these concepts. In addition, as also discussed in chapter 4, there is still much debate in the conceptual modeling literature regarding the meaning of these categories.

The theory of universals we propose is founded in a number of results in the literature of philosophy of language and descriptive metaphysics and supported by substantial empirical evidence from research in cognitive psychology. By using a number of formally defined meta-properties

(identity supply, identity carry, rigidity, relational dependency) we can generate a typology of universals, in which modeling concepts like types, roles, phases and mixins (among others) can be defined. Moreover, by considering the postulates proposed by this theory, we can formally define constraints on the possible relations to be established between these modeling concepts.

This theory represents an important contribution to the theory of conceptual modeling. First, because it can be used to formally characterize these concepts which are ubiquitous in conceptual modeling, and to provide methodological guidelines for how to apply these modeling concepts in practice, e.g., by helping the user to choose the most suitable concept to model a given general term in the universe of discourse. In addition, the theory can serve as a well-founded basis to harmonize different conceptions of these entities found in the literature.

Second, the theory can be used to evaluate the conceptual quality of type hierarchies (or concept taxonomies), and to solve recurrent modeling problems in the practice of conceptual modeling (as demonstrated in chapter 4). In particular, as discussed in section 9.7 below, by employing the categories and postulates of this theory we have been able to propose a *design pattern* capturing a solution to a recurrent and much discussed problem in *role modeling*.

9.4 Two Systems of Modal Logic with Sortal Quantification

In order to formally characterize the ontological distinctions and postulates proposed by the theory of universals aforementioned, we have presented two different (albeit complementary) extensions to traditional systems of quantified modal logics, namely, the languages L_1 and L_2 discussed in chapter 4. Although far from complete, these languages are successful in formally characterizing in a simpler way the important distinction between sortals and general property universals (non-sortals) w.r.t. to the former's exclusive ability to supply a principle of persistence and transworld identity to its instances. By doing this, we can address the limitations of classical (unrestricted extensional) modal logics, which reduces ontologically very different categories to the same logical footing.

In particular, in the case of L_1 we have employed the notion of *individual concept*, as a possible ontological interpretation to the notion of object identifiers in conceptual modeling. Moreover, as demonstrated by (Heller et al., 2004), a language such as L_1 can play an important role in

relating *endurantistic* and *perdurantistic* views of entities (views of an entity as an object and as a process, respectively).

The language L_1 is a system of *intentional modal logic* based on the first of the four systems proposed in (Gupta, 1980). However, it contains some important differences. In one sense, it can be seen as an extension of Gupta's proposal by elaborating on different types of sortal universals. However, differently from Gupta, it assumes an *absolute qualified view of identity* (van Leeuwen, 1991), as opposed to a *contingent view*. Thus, in our approach, although all principle of identities are sortal supplied, if two objects are identical according to one principle, then they are necessarily identical. For Gupta, conversely, two objects can be identical in a given world and diverse in other. The disadvantage of this contingent view of identity is that it denies the so-called *Leibniz's Law*, which is generally accepted as an axiom for the identity relation. As a result, it equates the identity relation with any equivalence relation.

9.5 Theory of Conceptual Part-Whole Relations

Part-whole relations are fundamental concepts both from a cognitive perspective, for the realization of many important cognitive tasks (Tversky, 1989), and from an ontological perspective, serving as a foundational for the formalization of other entities that compose a foundational ontology.

Parthood has been represented in practically all conceptual/object-oriented modeling languages (e.g., OML, UML, EER, LINGO), and although it has not yet been adopted as a modeling primitive in the Semantic web languages, some authors have already pointed out its relevance for reasoning in description logics (e.g., Lambrix, 2000).

Nonetheless, in conceptual modeling, the concepts of part and whole are often understood only superficially. Consequently, the representation of these concepts in conceptual modeling languages is based on the very minimal axiomatization that these notions require. Moreover, despite it being an active topic in the conceptual modeling literature, there is still much disagreement on what characterizes this relation and about the properties that part-whole relations should have from a conceptual point of view.

In chapter 5 we propose an ontological well-founded theory of conceptual part-whole relations, which aims at providing precise formal characterizations and real-world semantics for these concepts. This theory is organized in a typology of parthood relations. The categories in this typology are generated in the following manner. First, we consider a number of theories of parts from formal ontology in philosophy (Mereologies). These formal theories provide an important starting point

for the understanding and axiomatization of the notion of part. However, despite their importance, these theories concentrate merely on a formal notion of part and, hence, they fall short in fully characterizing the relation of parthood from a conceptual point of view. Thus as an extension to the mereological core of our typology, on the basis of the literature on meronymic relations in linguistic and cognitive sciences, we recognize the existence of not one but four types of part-whole relations, based on the type on ontological entities they relate. As a second dimension to create categories in this typology, we consider a number of meta-properties that part-whole can possess. These meta-properties designate: (i) whether objects can share parts; (ii) whether an object *only exists* being part of a specific whole (or of a whole of certain kind); (iii) whether an object *only exists* having a specific object as part (or a part of a specific kind). All these meta-properties are formally characterized using a modal logics approach.

The theory also shows that these two dimensions are not completely orthogonal, i.e., by selecting the type of a parthood relation based on the first dimension (i.e., based on the ontological category of its relata), a number of meta-properties on the second dimension are implied. In addition, each type of parthood selected in the first dimension determines a particular type of mereology (e.g., minimal, extensional) that should constitute the basic axioms of that specific relation.

This typology contributes to the theory of conceptual modeling by providing ontological foundations and real-world semantics for this important construct of conceptual modeling languages. Moreover, it helps to precisely characterize the relation between some of the entities forming this typology and one of the most important modeling concepts in conceptual modeling, namely, the concept of *roles* (see chapter 7). In particular, we use the distinction between *de re* and *de dicto* modality in the literature of philosophical logic to characterize the different formal properties of the relations of *specific dependent parthood* from the whole to the part, depending whether the whole universal is rigid or anti-rigid.

The explicit consideration of the modal meta-properties of part-whole relations together with those of the universals instantiated by their relata, constitute an important contribution not only to conceptual modeling, but also to the areas of software design and implementation. For example, inseparability and essentiality of parts in a part-whole relation constitute a sort of existential dependence relation, which imposes constraints on the relation between the life cycles of the relata. Understanding and making explicit these constraints is of fundamental importance not only for understanding and describing reality, but also, for example, for correctly specifying the relation between the life cycles of objects in object-oriented programming.

9.6 Contributions to the Problem of Transitivity in Conceptual Part-Whole Relations

The problem of transitivity of part-whole relations is a much debated topic not only in conceptual modeling but also in the linguistic and cognitive science literatures.

In many conceptual modeling languages (e.g., UML) part-whole relations are always considered transitive. However, as discussed in chapter 5, examples of fallacious cases of transitivity among part-whole relations abound.

In chapter 5, we discuss the relation between transitivity and the *unifying relations* characterizing the parts of a given whole. We conclude that if we consider a unique general sense of parthood, transitivity cannot be said to hold unrestricted, but only with respect to certain *contexts*. The delimitation of contexts, however, typically requires extensive knowledge of the domain being modeled.

In order to provide methodological assistance to the conceptual modeler in this task, we build on the different types of part-whole relations comprising the typology discussed in section 9.5. We demonstrate in chapter 5 that, by selecting the type of a part-whole relation in the first dimension aforementioned (i.e., according to the ontology category of its *relata*), we can determine which combination of parthood relations is transitive. The four types of part-whole relations considered in this respect are: (a) *subQuantityOf*; (b) *subCollectionOf*; (c) *memberOf*; (d) *componentOf*. As we have demonstrated, the combinations (a) + (a), (b) + (b) and (c) + (b) are always transitive. Thus, chains such as (a) + ... + (a), or (c) + (b) + ... + (b) will always delineate contexts. The combinations (c) + (c), (d) + (c), in contrast, are never transitive. Since most individuals represented in conceptual models are functional complexes, most part-whole relations represented are of the sort (d). This combination (d) + (d) is however itself non-transitive, and holds only in certain contexts. Other combinations such as (a) + (d), (b) + (d) are not possible. In these cases, the involved relations require *relata* of incompatible types. For instance, the combination *x is a subQuantity of y*, and *y is a componentOf z* is not possible since *y* cannot be both a quantity and a functional complex.

The precise definition of these contexts for functional parthood relations is of great importance not only for problem-solving in conceptual modeling, but also for software design and implementation. As discussed in (Odell & Bock, 1998), the transitivity of parthood relations has a strong impact on the propagation of properties and method invocations in object-oriented languages. For this reason, by building on the pioneering theory of transitivity of linguistic functional parthood relations proposed by (Vieu &

Aurnague, 2005), we propose a number of visual patterns that can be used to identify these contexts in conceptual class diagrams (chapter 7).

9.7 The *Role Modeling with Disjoint Admissible Types* Design Pattern

A known and much discussed problem in role modeling in the literature is the problem of specifying *admissible types* for Roles that can be filled by instances of disjoint types. This problem led (Steimann, 2000b), for example, to propose a complete separation of role and type hierarchies in conceptual models. As discussed in chapter 4, this solution implies a radical transformation to the metamodels of most of the current conceptual modeling languages.

By employing the theory of universals discussed in section 9.3, we propose an ontological design pattern capturing a standard solution to this problem. The adequacy of this design pattern is demonstrated by several examples throughout the thesis.

We believe that *ontological design patterns* such as this one, which captures standard solutions for recurrent conceptual modeling problems, represent an important contribution to the task of defining sound *engineering tools and principles* for the practice of conceptual modeling.

9.8 A Foundation for Harmonizing Different Concepts of Roles in the Literature

In (Wieringa et al., 1995), the authors offer an interesting alternative to the traditional notion of roles in conceptual and object-oriented modeling. They also propose a separation of role and type hierarchies, but for radically different reasons. Their main motivation is based on a philosophical problem known as *The Counting Problem* (Gupta, 1980). As we have shown in chapter 7, this problem is actually fallacious and, thus, the separation of role and type hierarchies cannot be argued for on this basis. Nonetheless, there is an important truth highlighted by their argument which is generally neglected in most conceptual modelling approaches, namely, that in different situations one might want to count “role instances” in different senses.

The notion of role proposed by Wieringa et al. is one in which role individuals are entities that are *existentially dependent* of their players. Moreover, role universals are responsible for supplying principles of identity

for their instances, which are different from the ones supplied by the universals of their players. This view is also shared by (Loebe, 2003).

By relying of the ontological categories of *moments* and *qua individuals* discussed in chapter 6, we manage to provide an ontological interpretation for the notion of roles proposed by Wieringa et al. and Loebe. Moreover, we manage to harmonize it with the more common view of roles taken in the literature (and, in particular, in the typology of section 9.3), and the one which more naturally represents the commonsense use of roles in ordinary language, namely, the conception of roles as *relationally dependent* and *anti-rigid substantial universals*.

Finally, by explicitly representing roles as both substantial universals and *qua individual* universals, we can account in an unambiguous way for the alternative senses of counting “role instances” previously mentioned.

9.9 An Aristotelian Ontology for Attributes, Weak Entities and Relationships

In chapter 6, we present the core of the foundational ontology proposed here, organized in terms of the so-called *Aristotelian ontological square* comprising the category pairs *Substantial-Substantial Universal*, *Moment-Moment Universal*. This ontology constitutes an important contribution to the theory of conceptual modeling, by enabling simple, unambiguous and ontologically interesting real-world semantics to be defined for the conceptual modeling categories of *attributes*, *weak entities* and *relationships*.

From a metaphysical point of view, these categories allow for the construction of a parsimonious ontology, based on the primitive and formally defined notion of *existential dependency*. This has been possible since, (i) by using this existential dependency meta-property we can differentiate which particulars in the domain are *substantials* and which are *moments*; (ii) based on the multiplicity of entities a moment depends on, we can distinguish between *qualities* and *relators*; (iii) with qualities we can explain *formal relations* and *weak entities*, and with relators, *material relations*; (iv) finally, with moments we can explain the relation of *exemplification*, and with exemplification *characterization*.

From a conceptual point of view, this ontology allows for the creation of a conceptual modeling language with great simplicity and preciseness of its constructs. For example, in the UML profile generated in chapter 8 all represented universals are either universals of independent entities (substantial universals), or universals which instances are existentially dependent of other instances (moment universals). In addition, all material associations can be reduced to primitive relations of existential dependence.

As a consequence, the user of this modeling language is guided in the choice of how to model the elements of the universe of discourse by a formally defined and, thus, unambiguous relation of existential dependence.

As discussed in chapter 6, despite being ubiquitous in conceptual modeling languages, relationships are particularly problematic and ambiguous constructs. However, by employing (explicitly represented) relators, we can provide not only an ontologically well-founded interpretation for them, but also one that can accommodate more subtle linguistic distinctions. Additionally, from a conceptual point of view, we can produce models that are free of cardinality constraint ambiguities, i.e., models in which *single-tuple* and *multiple-tuple* cardinality constraints are explicitly differentiated and represented. Furthermore, with the notions of qualities and relators we can define *qua individuals*, and with the latter we have managed to provide a foundation and terminological clarification, which harmonizes two competing notions of roles present in the conceptual modeling literature (see section 9.8). Furthermore, the ontological category of relators shall play an important role in extending our ontology with: (i) *an ontology of perdurants*: relators are typically founded on individual *processes* or *events* and, as discussed in chapter 6, they can play an important role as a mechanism for consistency preservation between static and dynamic conceptual models; (ii) *an ontology of social and intentional entities*: many social bonds (e.g., commitments, claims) are types of relators (Guizzardi & Wagner, 2005a). Moreover, the notion of relators presented here is akin to the one of affordances in organizational semiotics (Stamper et al., 2000).

9.10 Ontological Foundations for Attributes Values and Attribute Value Spaces

Traditionally, in conceptual modeling, value domains are taken for granted. In general, in conceptual modeling languages, they are only superficially considered by taking primitive datatypes as representing familiar mathematical sets (e.g., natural, integer, real, Boolean) and the focus has been almost uniquely on mathematical specification techniques.

In chapter 6, besides the entities that constitute the four-categorical ontology aforementioned, we explicitly take into account the *conceptual measurement structures* in which particular qualities are perceived (and conceived). By employing the theory of *conceptual spaces*, we can provide an ontological interpretation for the conceptual modeling notion of *attribute values* and *attribute value domains*.

As discussed in chapter 6, many benefits arise from the use of *conceptual spaces* and related notions (e.g., *quality domains* and *quality dimensions*), as

theoretical tools for conceptualizing attribute values and value domains. Firstly, quality domains and the constraints relating different quality dimensions captured in its structure can provide a sound basis for the conceptual modeling representations of the associated structured datatypes, constraining the possible values that their fields can assume. In other words, whatever constraints should be specified for a datatype, they must reflect the geometrical structure of the quality domain underlying this datatype.

Moreover, as demonstrated in chapters 6 and 7, from the structure of a certain quality domain, one can derive meta-properties (e.g., reflexivity, asymmetry, transitivity) for the formal relations based on this domain. That is to say, explicitly represented characteristics of conceptual spaces can provide an ontological explanation for these meta-properties.

Other approaches to ontology-based conceptual modeling in the literature (e.g., Evermann & Wand, 2001b) propose that the conceptual modeling representations of *natural Kinds* (a type of *substance sortals*) should be accompanied by a constraint specification, which restricts the possible values that the attributes of a substantial thing can assume, i.e., which delineates the *lawful state space* of that kind. As discussed in chapter 6, a conceptual space associated with a substance sortal is the cognitive counterpart of the lawful state space of that substance sortal. However, differently from the *lawful state spaces*, the *conceptual spaces* approach acknowledges that *quality domains* can themselves be multidimensional, exhibiting a constrained substructure that can occur in the definition of conceptual spaces associated with possibly different substantial universals. For this reason, we believe that the explicit representation of quality domains (associated with quality universals) as *datatypes* not only provides a further degree of structuring on lawful state spaces, but it also allows for a potential reuse of specifications of a subset of its constraints.

Finally, the explicit account of conceptual spaces and their constituent domains and dimensions afford the conceptualization of alternative measurement structures for a given ontological entity. Furthermore, the possibility of defining transformations and projections between these explicitly represented structures facilitates the tasks of knowledge sharing and semantic interoperability.

9.11 An Ontologically Well-Founded UML version for Conceptual Modeling and Ontology Representation

The ontology constructed throughout chapters 4 to 7 of this thesis is aimed at providing ontological foundations for conceptual modeling concepts and real-world semantics for the related constructs. The choice of

concepts on which we focus in this ontology (types and their instances, type taxonomies, associations, attributes, attribute values and attribute value spaces, part-whole relations) was strongly influenced by their importance in the practice of conceptual modeling, but also by their generality as being represented in constructs of existing conceptual modeling languages.

In chapter 8, we apply this foundational ontology as a reference for analyzing the ontological appropriateness of the Unified Modeling Language (UML) for the purpose of conceptual modelling and ontology representation. This effort is meant first as a case study of the language evaluation and (re)design framework proposed here. However, due to the following reasons, the ontologically well-founded re-designed version of UML proposed represents itself an important research contribution of this thesis: (i) the current status of UML as *de facto* standard modeling language; (ii) the growing interest in its adoption as a language for conceptual modelling and ontology representation; (iii) the current scarcity of *foundational* (as opposed to *lightweight*) conceptual modelling languages.

We emphasize, however, that this foundational ontology is *independent* of particular modeling languages and, hence, it can in principle be used as a foundation for analyzing any structural conceptual modeling language (e.g., EER, OML, LINGO, OWL). For instance, in (Guizzardi & Wagner, 2005a), an extension of the foundational ontology proposed here has been employed to analyze the ontological semantics of some enterprise modeling languages and frameworks (e.g., the REA framework) and, in (Wagner & Taveter, 2004), as the ontological foundation of the AORML agent-modeling language.

By using the evaluation and (re)design framework proposed here, we have managed to demonstrate cases in which each of the properties of a suitable representation mapping is lacking in the current UML meta-model when interpreted in terms of our reference ontology. In other words, as a conceptual modelling and ontology representation language, the UML meta-model can be shown to contain cases of construct *incompleteness*, *overload*, *redundancy* and *excess*.

In order to remedy this situation, we have proposed a number of modifications to the UML meta-model, producing a conceptually cleaner, semantically unambiguous and ontologically well-founded version of the language. Moreover, since the constructs in the re-designed meta-model can be unambiguously identified by formally defined ontological meta-properties, this approach provides the important additional benefit of methodologically supporting the user of the language in deciding how to model the elements of the universe of discourse.

We believe that one should attempt to shield as much as possible the user of a conceptual modelling language from the complexity of the underlying ontological theory. Therefore, it is our strategy to (whenever

possible) represent the ontological principles underlying a language in terms of *syntactical constraints* of this language. For example, since the *postulate 4.2* of our typology of universals (see section 9.3) is represented as a constraint in the redesigned UML meta-model, a user's model in which a class regarded as anti-rigid (e.g., customer) is a subclass of one regarded as rigid (e.g., person) is a *syntactically (grammatically) incorrect model*. Therefore, even if a conceptual modeller is unlearned about the ontological foundations of a given language, he can still produce only ontologically correct models (*intended models*) just by being accurate in the use of that language's grammar.

Finally, we emphasize that this strategy follows an analogous process occurring in human natural languages, since, according to (Chomsky, 1986), the grammars of natural languages have the properties they do to reflect the properties of ontological categories underpinning human cognition. Thus, as exemplified in chapter 4, some natural language constructions are ungrammatical for ontological reasons (e.g., only general terms representing sortal universals can be combined with quantifiers and determiners).

9.12 A Case Study on the Integration and Analysis of Semantic Web Ontologies

In chapter 3, we discuss the two-level approach defended in this thesis for semantic interoperation of conceptual models. We defend that in a first phase of off-line meaning negotiation, an expressive and ontologically adequate modelling language should be used for comparing the candidate models and making explicit their underlying ontological commitments. Once this meaning negotiation and semantic interoperation phase is complete, then a computationally efficient representation language can be used to express the results of this phase.

In chapter 8, we exemplify the first phase of this process by using the UML version proposed here to analyze and integrate several Semantic web lightweight ontologies. The scenario analyzed is on the integration of Semantic web ontologies in the scope of a context-aware service platform. This scenario was originally proposed in (Ríos, 2003) to illustrate the inadequacy of Semantic web languages in making explicit the underlying ontological commitments of the conceptualizations involved. Here, this scenario has been extended by considering real Semantic web models, which are structurally similar to the ones proposed by Ríos, and which are used by practitioners in concrete Semantic web applications.

As shown in chapter 8, through this case study, we have managed to demonstrate the adequacy of the language proposed and, consequently, of its underlying foundational ontology. In particular, this case study exemplifies the importance of the results achieved here to Semantic web-related projects such as A-MUSE⁸⁰ (Architecture Modelling Utility for Service Enabling) and AWARENESS⁸¹ (AWARE Mobile Network and ServiceS).

Finally, in chapter 2 we have illustrated the design of a domain-specific modelling language based on a domain ontology of geneology. In that chapter, this example is presented as an illustration of the language evaluation and (re)design framework proposed. However, as discussed there, the derivation of suitable syntactical and pragmatic constraints for the designed language depends on suitability of the domain ontology represented. Therefore, that example can now be seen also as an additional exemplification of the usefulness of the modelling language and the ontological concepts proposed here.

9.13 Future Work

In this thesis, we aim at developing ontological foundations for structural conceptual modeling concepts. As a result, the foundational ontology proposed is centered in the ontological category of endurants and endurant universals. In (Guizzardi & Wagner, 2004; 2005a; 2005b), we present an extension of this ontology as a component of a larger one named UFO (Unified Foundational Ontology).

UFO is organized in three incrementally layered *compliance sets*:

1. UFO-A: essentially the foundational ontology proposed here;
2. UFO-B: defines, as an increment to UFO-A, terms related to perdurants;
3. UFO-C: defines, as an increment to UFO-B, terms explicitly related to the spheres of intentional and social things, including linguistic things.

This division reflects a certain stratification of our “world”. It also reflects different degrees of scientific consensus: there is more consensus about the ontology of endurants than about the ontology of perdurants, and there is more consensus about the ontology of perdurants than about the ontology of intentional and social things.

⁸⁰ <http://a-muse.freeband.nl/>

⁸¹ <http://awareness.freeband.nl/>

The usefulness of this extended ontology has been already demonstrated in analyzing agent related concepts (Guizzardi & Wagner, 2004; 2005b) and enterprise modeling languages (Guizzardi & Wagner, 2005a). Nevertheless, UFO-B and UFO-C are still in a preliminary stage of development and, thus, substantial work still needs to be done in understanding and formally characterizing the entities constituting these ontologies.

As discussed in chapter 3, in domain engineering, ontology representation languages can play an important role, providing useful abstractions for representing domain models. However, once a domain model (domain ontology) is represented, there is still the problem of systematically deriving from it reusable artifacts (e.g., frameworks), which preserve the characteristics of the conceptualization expressed in that model. The same situation occurs in traditional software engineering, in which design and implementation specifications should also preserve the real-world semantics of the domain conceptualization (captured in a conceptual model of the domain). This problem is known as the *impedance mismatch problem* (Guizzardi & Falbo & Pereira Filho, 2001a).

In (Guizzardi & Falbo & Pereira Filho, 2001a,b; 2002) and (Falbo & Guizzardi & Duarte, 2002), we propose a systematic and semantics-preserving approach for deriving object-oriented frameworks from domain ontologies expressed in LINGO. The derivation methodology proposed comprises a spectrum of techniques, namely, mapping directives, design patterns and formal translation rules. For example, in (Guizzardi & Falbo & Pereira Filho, 2002) a design pattern is proposed that guarantees the preservation of some ontological properties of part-whole relations (irreflexivity, asymmetry, transitivity and shareability) in object-oriented implementations.

When evaluated in terms of the foundational ontology proposed here, LINGO can be shown to be seriously incomplete. A natural extension of the work aforementioned is the investigation of similar methodological mechanisms for deriving object-oriented implementations and frameworks that preserve the ontological properties of domain models expressed in the conceptual modeling language proposed here.

There are also interesting possibilities for future work on the topic of domain-specific visual languages evaluation and (re)design. Firstly, a more systematic account is needed of the connection between ontological meta-categories (and their meta-properties) and, systems of visual signs. An interesting approach to pursue this topic would be in the lines of the *Algebraic Semiotics* program proposed in (Goguen, 1999), since such an approach should consider a more comprehensive exploration of semiotic theory, but doing it in a precise way. However, differently from the latter, domain models should be suitably represented in terms of expressive

domain ontologies, in which ontological meta-properties of the elements of a conceptualization are explicitly represented, as opposed to focusing solely on abstract (algebraic and category-theoretical) domain specifications.

The establishment of systematic connections between real-world semantic domains, meta-models, and visual systems of concrete syntaxes shall provide a solid basis for the construction of semantic aware (meta) modeling case tools for domain-specific visual language evaluation and (re)design. Additionally, with the development of systematic approaches for deriving frameworks from expressive domain ontologies (along the lines previously discussed) these tools can also support the creation of libraries of reusable domain artifacts, thus, bridging the gap between domain-specific conceptual modeling and design/implementation. One should refer to (Falbo et al., 2002b; Falbo et al., 2003) for an example of an ontology-based semantic software engineering environment based on the frameworks derivation approach aforementioned.

In this thesis, we have shown many conceptual modeling benefits resulting from the application of the *theory of conceptual spaces* in the foundation ontology proposed here. However, there are many other aspects of this theory, which were not explored in this work, but which shall be pursued in future investigations.

For example, as briefly discussed in chapter 6, conceptual spaces can be used for the construction of *Portions* (Gerlst & Pribbenow, 1995), i.e., dynamically assembled collectives built by selecting certain parts of an integral object, according to certain internal properties of this object. Examples include: (a) the reddish parts of an object; (b) the tall animals in the group. The notion of *contrast classes* in the theory of conceptual spaces (Gardenfors 2000; 2004) can allow for interesting cases of non-monotomic reasoning in situations such as depicted in (b). For instance, although every squirrel is an animal, a tall squirrel is not a tall animal. As argued by (Gerlst & Pribbenow, 1995), *Portions* can play an important role in reasoning about physical objects. In particular, they can be useful in *context-dependent reasoning*, by supporting the construction of dynamically created aggregates depending on a given context (e.g., the crowded areas of a geographical location, the restored parts of a museum).

Another aspect of this theory that could be useful in context-aware scenarios is the notion of *context matrices*, which can be used to emphasize certain quality dimensions of conceptual spaces in detriment of others depending on the circumstance. Take, for instance, a *wayfinding service*. In a *day context*, the best wayfinding landmark to be used by this service could be a façade with the most contrasting color. In contrast, in a *night context*, it can be simply the highest or widest landmark (Raubal, 2004).

Finally, conceptual spaces provide a sound mechanism for accounting for relations such as *resemblance*, *complementation*, and *opposition* between

universals. As a consequence, one can also account for relations such as these between individuals instantiating those universals. Besides the theoretical importance of this mechanism for a theory of conceptual modeling (as discussed in chapter 6), it allows for the construction of conceptual models which can support approximate (inexact) reasoning applications (e.g., *inexact service matching*). This topic should also be explored in future research.

Besides contributing to the development of a philosophically and cognitive principled theory of conceptual modeling, it is also a general objective of this thesis to work towards the development of sound practical tools for the disciplines of conceptual modeling and ontological engineering. We believe that the development of these tools constitutes an important step, contributing to the maturity of these disciplines as engineering disciplines.

In this thesis, we have pursue this latter goal in several manners, for instance, by proposing a systematic evaluation and (re)design method, by providing methodological guidelines for helping the user in deciding how to model elements of given conceptualization, by proposing a modeling language in which ontological principles are incorporated as the language's syntax constraints, or by proposing formal languages in which these ontological principles are embedded in the language's semantic theory. Moreover, an important contribution of this thesis in this direction is the creation of an *ontological design pattern* that captures a situation-independent solution for a classical and recurrent problem in conceptual modeling. With the extension of the ontological theory proposed here to incorporate other aspects of reality (UFO-B and UFO-C), we shall continue to investigate the development of conceptual modeling engineering tools in all these directions.

